



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2010
HIGHER SCHOOL CERTIFICATE
TRIAL PAPER

Mathematics Extension 2

General Instructions

- Reading Time – 5 Minutes
- Working time – 3 Hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Each Question is to be returned in a separate bundle.
- All necessary working should be shown in every question.
- Answer in simplest exact form unless otherwise instructed.

Total Marks – 120

- Attempt questions 1 – 8

Examiner: *A.M.Gainford*

- NOTE: This is a trial paper only and does not necessarily reflect the content or format of the final Higher School Certificate examination paper for this subject.

Question 1. (15 marks) **(Start a new answer sheet.)**

Marks

(a) Evaluate $\int_0^3 \frac{x}{\sqrt{x^2+16}} dx.$ **2**

(b) Find $\int (\cos^2 x - \sin^2 x) dx.$ **1**

(c) Use integration by parts to find **2**

$$\int x e^{-x} dx.$$

(d) (i) Find real numbers a and b such that **2**

$$\frac{1-3x}{x^2-3x+2} = \frac{a}{x-1} + \frac{b}{x-2}.$$

(ii) Hence find **2**

$$\int \frac{1-3x}{x^2-3x+2} dx.$$

(e) Evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{\cos x + 2} dx.$ **2**

(f) (i) If $I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx, n = 1, 2, 3, \dots$ **2**

show that $I_n + I_{n-2} = \frac{1}{n-1}, n = 2, 3, 4, \dots$

(ii) Hence evaluate **2**

$$\int_0^{\frac{\pi}{4}} \tan^5 x dx.$$

Question 2. (15 marks) (Start a new answer sheet.)

Marks

- (a) If $u = 3 - 4i$ and $v = 2 - 2i$ find

4

(i) $u\bar{v}$

(ii) $\sqrt{u}$

(iii) v in modulus-argument form.

(iv) v^4 using De Moivre's theorem.

- (b) On an Argand diagram shade the region that is satisfied by both the conditions

2

$$3 \leq |z - 4i| \leq 4 \text{ and } -\frac{\pi}{4} < \arg(z - 4i) < \frac{\pi}{4}$$

- (c) Sketch, on separate Argand diagrams, the locus of the complex number z satisfying

4

(i) $z^2 - (\bar{z})^2 = i$

(ii) $|z - 1| = \operatorname{Re}(z)$

- (d) It is given that $z = \cos \theta + i \sin \theta$ where $0 < \arg z < \frac{\pi}{2}$.

5

(i) Show that $z + 1 = 2 \cos \frac{\theta}{2} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right)$ and express $z - 1$ in modulus-argument form.

(ii) Hence show that $\operatorname{Re} \left(\frac{z - 1}{z + 1} \right) = 0$.

Question 3. (15 marks) (Start a new answer sheet.)

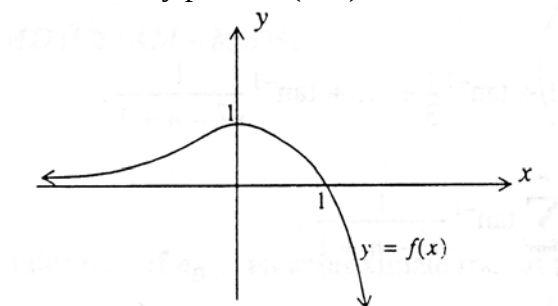
Marks

- (a) (i) Show that $z = 1 + i$ is a root of $z^2 - (3 - 2i)z + (5 - i) = 0$. **3**
- (ii) Find the other root of the equation.
- (b) If α , β and γ are roots of the equation $x^3 + qx - 2 = 0$ find, in terms of q , **3**
the monic cubic polynomial equation whose roots are α^2 , β^2 and γ^2 .
- (c) (i) Use De Moivre's theorem to find $\cos 5\theta$ in terms of powers of $\cos \theta$. **6**
- (ii) Use the result in (i) to solve the equation
- $$16x^4 - 20x^2 + 5 = 0$$
- (d) If ω represents one of the complex roots of the equation $z^3 - 1 = 0$
- (i) Show that $1 + \omega + \omega^2 = 0$.
- (ii) Evaluate $(1 - \omega^8)(1 - \omega^4)(1 - \omega^2)(1 - \omega)$.

Question 4 (15 marks) **(Start a new answer sheet.)**

- (a) The graph of $y = f(x)$ is sketched below.

There is a stationary point at $(0, 1)$.



Use this graph to sketch the following, on separate diagrams, showing essential features.

(i) $y = f\left(\frac{x}{2}\right)$

2

(ii) $y = x + f(x)$

2

(iii) $y = \frac{1}{f(x)}$

2

(iv) $y = f\left(\frac{1}{x}\right)$

2

(b) (i) Find $\int \frac{1}{x^2 \sqrt{9-x^2}} dx$, using the substitution $x = 3\cos\theta$.

4

(ii) Evaluate $\int_1^e x^3 \log_e x dx$.

- (c) Given that the quartic polynomial $p(x) = x^4 - 5x^3 - 9x^2 + 81x - 108$ has a zero of multiplicity 3, factorise $p(x)$ completely, and find all its zeroes.

3

Question 5 (15 marks) (Start a new answer sheet.)

Marks

- (a) A particle is moving under gravity in a fluid which exerts a resistance to its motion, per unit mass, k times its speed (k is constant). **2**
- (i) If the particle falls vertically from rest, show that its terminal velocity is $V_T = \frac{g}{k}$, where g is acceleration due to gravity. **2**
- (ii) If the particle is projected vertically upward with velocity V_T show that after time t seconds **6**
- (α) its speed is $V_T (2e^{-kt} - 1)$
- (β) its height above the starting point is $\frac{1}{k} V_T (2 - 2e^{-kt} - kt)$
- (iii) Hence find an expression for the greatest height reached in terms of V_T and k . **2**
- (b) A box contains 6 white balls and 2 black balls. Balls are selected at random, one at a time, and not replaced. A note is kept of the number, X , of the draw which first yields a black ball. If this experiment is repeated many times, find: **5**
- (i) the most probable value of X ;
- (ii) the probability that $X > 4$.

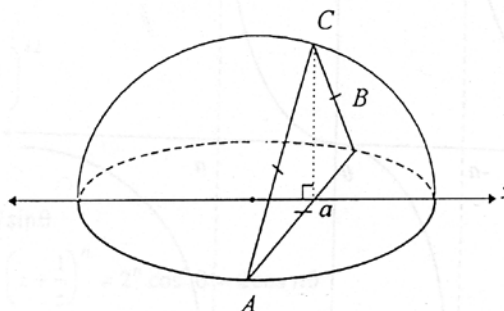
Question 6 (15 marks) (Start a new answer sheet.)

- (a) A council has 14 councillors: 6 Labor, 5 Liberal and 3 Independents. 6
Five councillors are chosen at random to form a committee.

- (i) (α) How many different committees can be formed?
- (β) Find the probability that the committee will have a majority of Labor councillors.
- (ii) (α) Show that the number of different committees which can be formed with at least one councillor from each of the groups Labor, Liberal, and Independent is 1365.
- (β) Given that the committee contains at least one councillor from each of the groups Labor, Liberal, and Independent, find the probability that the committee will have a majority of Labor councillors.

- (b) The circle $x^2 + y^2 = 4$ is rotated about the line $x = 5$ to form a torus. Use the method of cylindrical shells to prove that the volume of the solid is $40\pi^2$ cubic units. 4

- (c) The solid drawn at right has a circular base of radius 3 units in the horizontal plane. Vertical cross-sections perpendicular to the diameter along the x -axis are equilateral triangles.



- (i) A vertical slice of width Δa is positioned at the point where $x = a$. 3
If the volume of the slice is ΔV , show that $\Delta V = \sqrt{3}(9 - a^2)\Delta a$.
- (ii) Hence determine the volume of the solid. 2

Question 7 (15 marks) (Start a new answer sheet.)

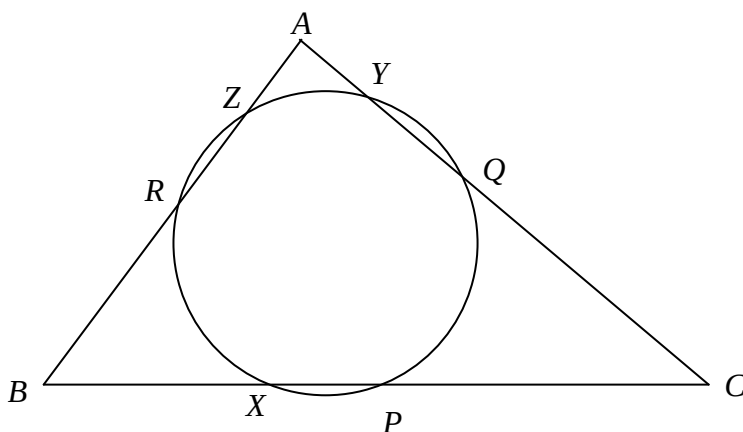
- (a) On polling day in Rock Island City the ratio of electoral votes in the only four polling booths A, B, C, and D was 5:4:3:8 respectively. The percentages of votes for Mr Jones in these booths were 60%, 50%, 40%, and 70% respectively. 4

- (i) Find the probability that a voter chosen at random voted for Mr Jones.
- (ii) If ten voters of this city were chosen at random, find the probability that Mr Jones gained
- (α) at least 8 votes
- (β) no more than 2 votes.

- (b) The equation $e^{2x} \log_e y = 3$ implicitly defines y as a function of x . 3

Find $\frac{dy}{dx}$ as a function of y .

- (c) 8



In the diagram above, P , Q , and R are the midpoints of the sides BC , CA , and AB respectively of a triangle ABC . The circle drawn through the points P , Q , and R meets the sides BC , CA , and AB again at X , Y , and Z respectively.

Copy the diagram to your answer sheet.

- (i) Briefly explain why $RPCQ$ is a parallelogram.
- (ii) Show that $\triangle XCQ$ is isosceles.
- (iii) Show that $AX \perp BC$.

Question 8 (15 marks) (Start a new answer sheet.)

- (a) Five women and four men are to be seated at a round table. 5
- (i) In how many ways may this be done without restrictions?
- (ii) In how many ways may this be done if no two men are to be seated together?
- (iii) If one man and one woman are a married couple, what is the probability that they are seated together, given the conditions of part (ii)?

- (b) One root of the equation $x^3 + ax^2 + bx + c = 0$ is equal to the sum of the other two roots. 4

Show that $a^3 - 4ab + 8c = 0$.

- (c) (i) Graph, in the same xy -plane, the curves 6

$$y = x^{-\frac{2}{3}}, x > 0 \quad \text{and} \quad y = (x-1)^{-\frac{2}{3}}, x > 1$$

- (ii) Hence, or otherwise, given the sum S , where

$$S = 1 + \frac{1}{\sqrt[3]{2^2}} + \frac{1}{\sqrt[3]{3^2}} + \dots + \frac{1}{\sqrt[3]{(10^9)^2}},$$

find the two consecutive integers between which the sum S lies.

This is the end of the paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Sydney Boys Extension 2 2010

Question 1

$$a) \int_0^3 \frac{x}{\sqrt{x^2+16}} dx$$

$$= \left[\sqrt{x^2+16} \right]_0^3$$

$$= 5 - 4$$

$$= \underline{\underline{1}}$$

$$b) \int (\cos^2 x - \sin^2 x) dx$$

$$= \int \cos 2x dx$$

$$= \underline{\underline{\frac{1}{2} \sin 2x + c}}$$

$$c) I = \int x e^{-x} dx$$

$$u = x$$

$$v = -e^{-x}$$

$$du = dx$$

$$dv = e^{-x} dx$$

$$I = -x e^{-x} + \int e^{-x} dx$$

$$= \underline{\underline{-x e^{-x} - e^{-x} + c}}$$

$$d)(i) a(x-2) + b(x-1) = 1-3x$$

$$\frac{x=2}{b=-5}$$

$$\frac{x=1}{-a=-2}$$

$$a=2$$

$$\therefore \underline{\underline{a=2, b=-5}}$$

$$(ii) \int \frac{1-3x}{x^2-3x+2} dx$$

$$= \int \left[\frac{2}{x-1} - \frac{5}{x-2} \right] dx$$

$$= \underline{\underline{2 \log(x-1) - 5 \log(x-2) + c}}$$

$$e) \int_0^{\frac{\pi}{2}} \frac{dx}{\cos x + 2}$$

$$t = \tan \frac{x}{2}$$

$$dx = \frac{2dt}{1+t^2}$$

$$= \int_0^1 \frac{2dt}{1+t^2 + 2}$$

$$x=0, t=0$$

$$x=\frac{\pi}{2}, t=1$$

$$= \int_0^1 \frac{2dt}{1+t^2+2+2t^2}$$

$$= \int_0^1 \frac{2dt}{3+t^2}$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1$$

$$= \underline{\underline{\frac{2}{\sqrt{3}} \left(\frac{\pi}{6} - 0 \right) = \frac{\pi}{3\sqrt{3}}}}$$

$$f) I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx$$

$$I_n + I_{n-2}$$

$$= \int_0^{\frac{\pi}{4}} (\tan^n x + \tan^{n-2} x) dx$$

$$= \int_0^{\frac{\pi}{4}} \tan^{n-2} x (\tan^2 x + 1) dx$$

$$= \int_0^{\frac{\pi}{4}} \tan^{n-2} x \sec^2 x dx$$

$$= \frac{1}{n-1} \left[\tan^{n-1} x \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{n-1} (1-0)$$

$$= \underline{\underline{\frac{1}{n-1}}}$$

$$(ii) \int_0^{\frac{\pi}{4}} \tan^5 x dx = I_5$$

$$= \frac{1}{4} - I_3$$

$$= \frac{1}{4} - \frac{1}{2} + I_1$$

$$= -\frac{1}{4} + \int_0^{\frac{\pi}{4}} \tan x dx$$

$$= -\frac{1}{4} - \left[\log \cos x \right]_0^{\frac{\pi}{4}}$$

$$= -\frac{1}{4} - \log \frac{1}{\sqrt{2}} + \log 1$$

$$= \underline{\underline{-\frac{1}{4} - \log \frac{1}{\sqrt{2}}}}$$

Question 2

$$a)(i) u \bar{v} = (3-4i)(2+2i)$$

$$= 6+6i-8i+8$$

$$= \underline{\underline{14-2i}}$$

$$(ii) \sqrt{u} = \sqrt{3-4i}$$

$$a^2 - b^2 = 3$$

$$2ab = -4$$

$$a^2 - \frac{4}{a^2} = 3$$

$$b = -\frac{2}{a}$$

$$a^4 - 3a^2 - 4 = 0$$

$$(a^2-4)(a^2+1) = 0$$

$$a^2 = 4 \text{ or } a^2 = -1$$

$$a = \pm 2 \text{ no real solutions}$$

$$\therefore \underline{\underline{\sqrt{u} = \pm(2-i)}}$$

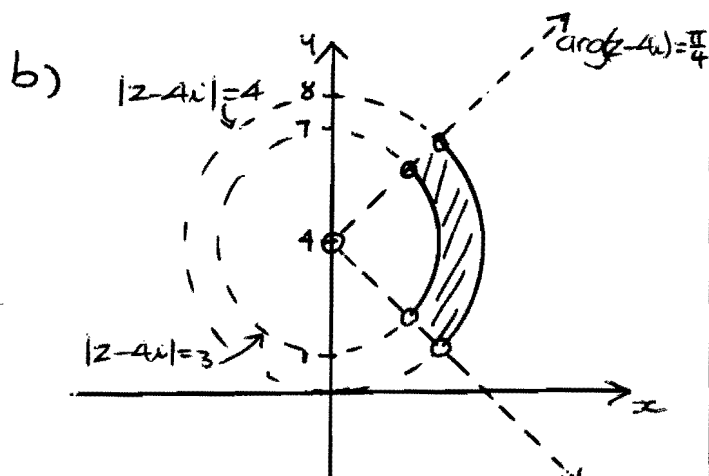
$$(iii) |v| = \sqrt{2^2+2^2} = 2\sqrt{2}$$

$$\arg v = \tan^{-1} \frac{-2}{2} = -\frac{\pi}{4}$$

$$\underline{\underline{v = 2\sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4} \right)}}$$

$$(iv) v^4 = (2\sqrt{2})^4 \operatorname{cis}(-\pi)$$

$$= \underline{\underline{-64}}$$



$$\therefore \operatorname{Re}\left(\frac{z-i}{z+i}\right) = 0$$

Question 3

a) $P(1+i)$

$$= (1+i)^2 - (3-2i)(1+i) + 5-i$$

$$= 1+2i-1-3-3i+2i-2+5-i$$

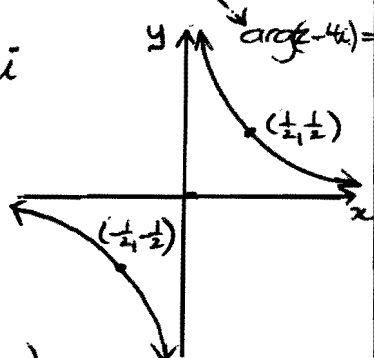
$$= 0$$

$\therefore 1+i$ is a root

c) (i) $z^2 - (\bar{z})^2 = i$

$$4ixy = i$$

$$xy = \frac{1}{4}$$



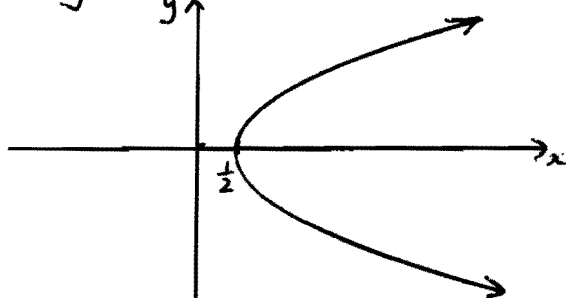
(ii) $|z-1| = \operatorname{Re}(z)$

$$(x-1)^2 + y^2 = x^2$$

$$x^2 - 2x + 1 + y^2 = x^2$$

$$y^2 = 2x - 1$$

$$y^2 = 2(x - \frac{1}{2})$$



d) $z+1$

$$= \cos\theta + 1 + i\sin\theta$$

$$= 2\cos^2\frac{\theta}{2} - 1 + 1 + 2i\sin\frac{\theta}{2}\cos\frac{\theta}{2}$$

$$= 2\cos\frac{\theta}{2} \left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2} \right)$$

$$z-1$$

$$= \cos\theta - 1 + i\sin\theta$$

$$= 1 - 2\sin^2\frac{\theta}{2} - 1 + 2i\sin\frac{\theta}{2}\cos\frac{\theta}{2}$$

$$= -2\sin\frac{\theta}{2} \left(\sin\frac{\theta}{2} - i\cos\frac{\theta}{2} \right)$$

$$\frac{z-1}{z+1} = \frac{-2\sin\frac{\theta}{2}(\sin\frac{\theta}{2} - i\cos\frac{\theta}{2})}{2\cos\frac{\theta}{2}(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2})} \times \frac{\cos\frac{\theta}{2} - i\sin\frac{\theta}{2}}{\cos\frac{\theta}{2} - i\sin\frac{\theta}{2}}$$

$$= \frac{-\sin\frac{\theta}{2}(\sin\frac{\theta}{2}\cos\frac{\theta}{2} - i\sin^2\frac{\theta}{2} - i\cos^2\frac{\theta}{2} - \sin\frac{\theta}{2}\cos\frac{\theta}{2})}{\cos\frac{\theta}{2}(\cos^2\frac{\theta}{2} + \sin^2\frac{\theta}{2})}$$

$$= \frac{\sin\frac{\theta}{2}(\sin^2\frac{\theta}{2} + \cos^2\frac{\theta}{2})}{\cos\frac{\theta}{2}}$$

$$= i \tan\frac{\theta}{2}$$

b) let $y = x^2 \Rightarrow x = y^{\frac{1}{2}}$

$$y^{\frac{3}{2}} + qy^{\frac{1}{2}} - 2 = 0$$

$$y^{\frac{1}{2}}(y+q) = 2$$

$$y(y^2 + 2qy + q^2) = 2$$

$$y^3 + 2qy^2 + q^2y - 2 = 0$$

c) $\cos 5\theta = (\cos\theta)^5$

$$= \cos^5\theta - 5\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta - \sin^5\theta$$

equating real parts

$$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$$

$$= \cos^5\theta - 10\cos^3\theta(1-\cos^2\theta) + 5\cos\theta(1-\cos^2\theta)^2$$

$$= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta$$

$$= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$

(ii) $16x^4 - 20x^2 + 5 = 0$

let $x = \cos\theta$

$$\frac{\cos 5\theta}{\cos\theta} = 0$$

$$\cos 5\theta = 0, \cos\theta \neq 0$$

$$5\theta = \pm\pi k, k = 1, 2$$

$$\theta = \pm\frac{\pi}{5}, \pm\frac{2\pi}{5}$$

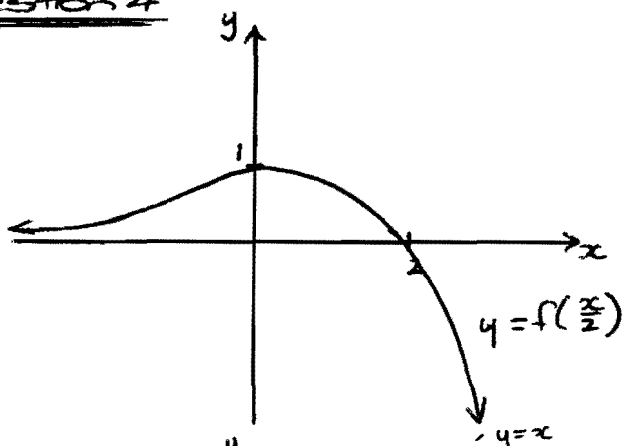
$$x = \cos\frac{\pi}{5}, -\cos\frac{\pi}{5}, \cos\frac{2\pi}{5}, -\cos\frac{2\pi}{5}$$

d) i) $w^3 - 1 = 0$
 $(w-1)(w^2+w+1) = 0$
 $w=1$ or $\underline{w^2+w+1=0}$
 not a solution as w is complex root

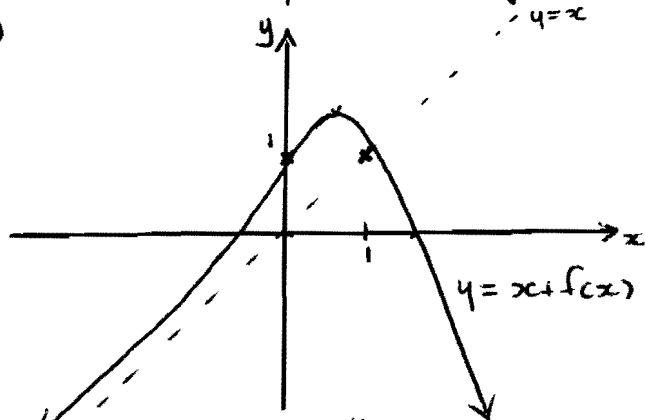
ii) $(1-w^8)(1-w^4)(1-w^2)(1-w)$
 $= (1-w^2)(1-w)(1-w^2)(1-w)$
 $= [(1-w^2)(1-w)]^2$
 $= (1-w-w^2+w^3)^2$
 $= (2-w-w^2)^2$
 $= (3-1-w-w^2)^2$
 $= 3^2$
 $= \underline{9}$

Question 4

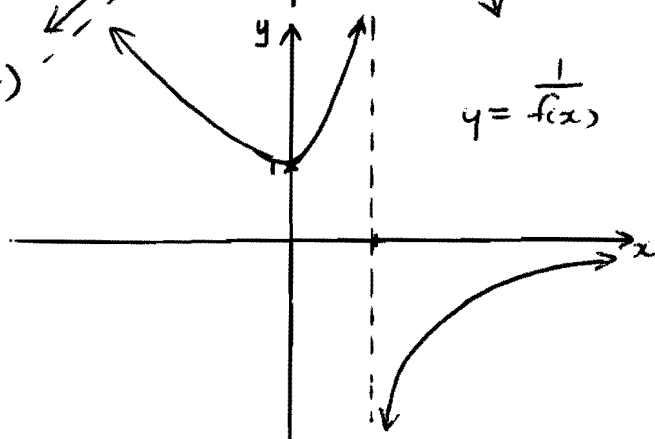
a)



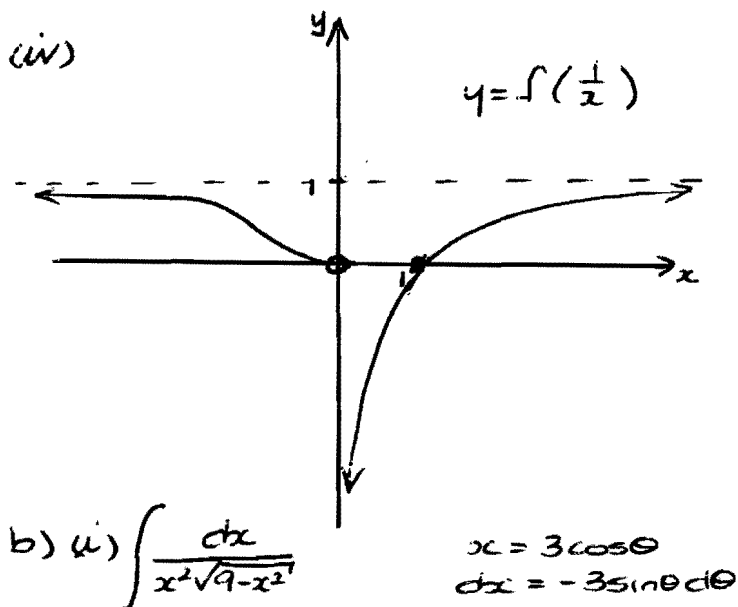
ii)



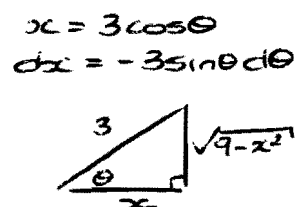
iii)



iv)



b) i) $\int \frac{dx}{x^2 \sqrt{9-x^2}}$
 $= \int \frac{-3 \sin \theta d\theta}{9 \cos^2 \theta \cdot 3 \sin \theta}$
 $= -\frac{1}{9} \int \sec^2 \theta d\theta$
 $= -\frac{1}{9} \tan \theta + C$
 $= -\frac{x}{9\sqrt{9-x^2}} + C$



ii) $\int_1^e x^3 \log x dx$

$u = \log x$ $v = \frac{1}{4}x^4$
 $du = \frac{dx}{x}$ $dv = x^3 dx$
 $= \left[\frac{1}{4}x^4 \log x \right]_1^e - \frac{1}{4} \int_1^e x^3 dx$
 $= \frac{1}{4}e^4 - \frac{1}{16} [x^4]_1^e$
 $= \frac{1}{4}e^4 - \frac{1}{16}e^4 + \frac{1}{16}$
 $= \underline{\underline{\frac{3}{16}e^4 + \frac{1}{16}}}$

c) $p(x) = x^4 - 5x^3 - 9x^2 + 81x - 108$

$p'(x) = 4x^3 - 15x^2 - 18x + 81$

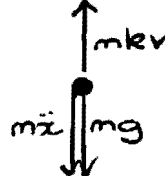
$p''(x) = 12x^2 - 30x - 18$
 $= 6(2x^2 - 5x - 3)$
 $= 6(2x+1)(x-3)$

$p(3) = p'(3) = p''(3) = 0$

$\therefore p(x) = \underline{\underline{(x-3)^3(x+4)}}$

zeros are 3, 3, 3 and -4

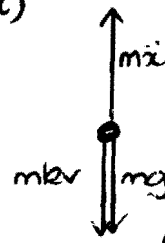
Question 5

a)  $m\ddot{x} = mg - mkv$
 $\ddot{x} = g - kv$

terminal velocity occurs when $\ddot{x} = 0$

$$\text{i.e. } g - kv = 0$$

$$\underline{v = \frac{g}{k}}$$

(ii)  $m\ddot{x} = -mg - mkv$
 $\ddot{x} = -g - kv$
 $\frac{dv}{dt} = -g - kv$
 $\int_0^t dt = -\int_{v_T}^v \frac{dv}{g + kv}$

$$t = \left[-\frac{1}{k} \log(g + kv) \right]_{v_T}^v$$

$$= -\frac{1}{k} \log \left(\frac{g + kv}{g + kv_T} \right)$$

$$-kt = \log \left(\frac{g + kv}{g + kv_T} \right)$$

$$e^{-kt} = \frac{g + kv}{g + kv_T}$$

$$= \frac{\frac{g}{k} + v}{\frac{g}{k} + v_T}$$

$$= \frac{v_T + v}{2v_T}$$

$$2v_T e^{-kt} = v_T + v$$

$$v = 2v_T e^{-kt} - v_T$$

$$= \underline{v_T (2e^{-kt} - 1)}$$

(b) $\frac{dx}{dt} = v_T (2e^{-kt} - 1)$

$$\int_0^x dx = v_T \int_0^t (2e^{-kt} - 1) dt$$

$$x = v_T \left[-\frac{2}{k} e^{-kt} - t \right]_0^t$$

$$= v_T \left(-\frac{2}{k} e^{-kt} - t + \frac{2}{k} \right)$$

$$= \underline{\frac{1}{k} v_T (2 - 2e^{-kt} - kt)}$$

(iii) greatest height occurs when $v = 0$

$$\text{i.e. } v_T (2e^{-kt} - 1) = 0$$

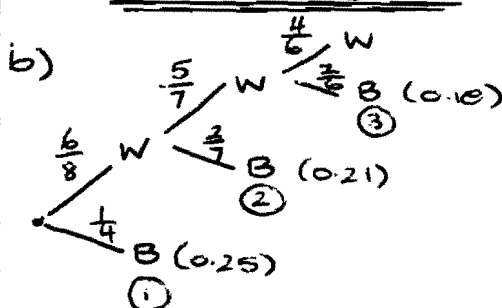
$$e^{-kt} = \frac{1}{2}$$

$$-kt = \log \frac{1}{2}$$

$$t = -\frac{1}{k} \log \frac{1}{2} = \frac{1}{k} \log 2$$

$$x = v_T \left(2 - 2\left(\frac{1}{2}\right) + \log \frac{1}{2} \right)$$

$$= \underline{v_T \left(1 + \log \frac{1}{2} \right)}$$



(i) The most probable value of X is 1

(ii) $P(X > 4) = P(\text{first 4 balls are W})$

$$= \frac{6}{8} \times \frac{5}{7} \times \frac{4}{6} \times \frac{3}{5}$$

$$= \underline{\frac{3}{14}}$$

Question 6

a) (i) Committees = ${}^{14}C_5$
 $= \underline{2002}$

(b) $P(\text{majority Labor})$

$$= P(3 \text{ Labor}) + P(4 \text{ Labor}) + P(5 \text{ Labor})$$

$$= {}^6C_3 \times {}^8C_2 + {}^6C_4 \times {}^8C_1 + {}^6C_5$$

$${}^{14}C_5$$

$$= \underline{\frac{49}{143}}$$

(ii) Committees

$$= {}^6C_3 \times {}^5C_1 \times {}^3C_1 + {}^6C_2 \left({}^5C_2 \times {}^3C_1 + {}^5C_1 \times {}^3C_2 \right)$$

$$+ {}^6C_1 \left({}^5C_3 \times {}^3C_1 + {}^5C_2 \times {}^3C_2 + {}^5C_1 \times {}^3C_3 \right)$$

$$= 300 + 675 + 390$$

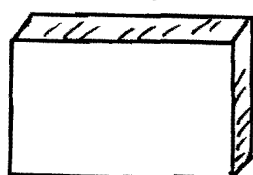
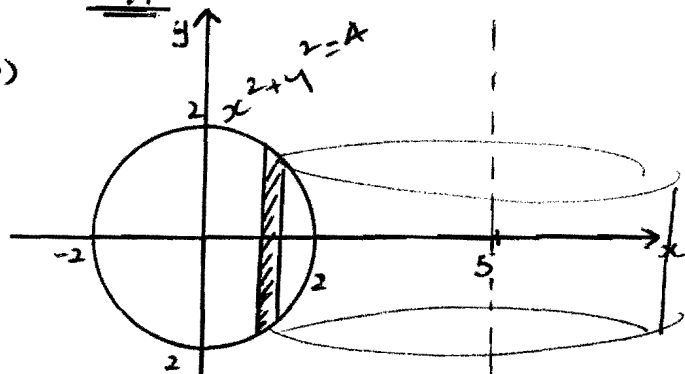
$$= \underline{1365}$$

(b) $P(\text{majority Labor})$

$$= \frac{{}^6C_3 \times {}^5C_1 \times {}^3C_1}{1365}$$

$$= \frac{20}{91}$$

b)



$$2y = 2\sqrt{4-x^2}$$

$$2\pi(5-x)\Delta x$$

$$A(x) = 2\pi(5-x) \cdot 2\sqrt{4-x^2}$$

$$\Delta V = 4\pi(5-x)\sqrt{4-x^2}\Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=-2}^2 4\pi(5-x)\sqrt{4-x^2}\Delta x$$

$$= 4\pi \int_{-2}^2 (5-x)\sqrt{4-x^2} dx$$

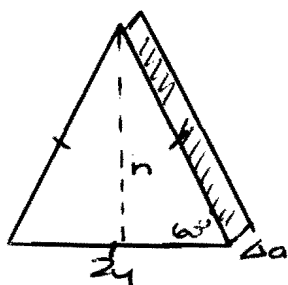
$$= 20\pi \int_{-2}^2 \sqrt{4-x^2} dx - 4\pi \int_{-2}^2 x\sqrt{4-x^2} dx$$

$$= 20\pi \times \frac{1}{2}\pi(2)^2 - 0$$

$$= 20\pi \times 2\pi$$

$$= \underline{40\pi^2 \text{ units}^3}$$

c)



$$\begin{aligned} \frac{h}{y} &= \tan 60^\circ \\ h &= y \tan 60^\circ \\ &= \sqrt{3}y \end{aligned}$$

$$\begin{aligned} A(a) &= \frac{1}{2} \times 2y \times \sqrt{3}y \\ &= \sqrt{3}y^2 \\ &= \sqrt{3}(\sqrt{9-a^2}) \end{aligned}$$

$$\underline{\Delta V = \sqrt{3}(\sqrt{9-a^2}) \Delta a}$$

$$\begin{aligned} \text{(ii)} V &= \lim_{\Delta x \rightarrow 0} \sum_{x=-3}^3 \sqrt{3}(\sqrt{9-x^2})\Delta x \\ &= \sqrt{3} \int_{-3}^3 \sqrt{9-x^2} dx \\ &= \sqrt{3} \times \frac{1}{2} \pi (3)^2 \\ &= \underline{\underline{\frac{9\sqrt{3}}{2} \pi \text{ units}^3}} \end{aligned}$$

Question 7

$$\begin{aligned} \text{a) (i)} P(\text{Jones}) &= \frac{0.6 \times 5 + 0.5 \times 4 + 0.4 \times 3 + 0.7 \times 8}{20} \\ &= \underline{\underline{\frac{5.9}{100}}} \end{aligned}$$

(ii) Let $X = \# \text{ votes for Jones}$

$$\begin{aligned} P(X \geq 8) &= P(X=8) + P(X=9) + P(X=10) \\ &= \binom{10}{8} \left(\frac{41}{100}\right)^2 \left(\frac{59}{100}\right)^8 + \binom{10}{9} \left(\frac{41}{100}\right) \left(\frac{59}{100}\right)^9 \\ &\quad + \left(\frac{59}{100}\right)^{10} \end{aligned}$$

$$= 0.1516993349$$

$$= \underline{0.1517} \text{ (to 4 dp)}$$

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= \left(\frac{41}{100}\right)^{10} + \binom{10}{1} \left(\frac{41}{100}\right)^9 \left(\frac{59}{100}\right) + \binom{10}{2} \left(\frac{41}{100}\right)^8 \left(\frac{59}{100}\right)^2$$

$$= 0.01457376613$$

$$= \underline{0.0146} \text{ (to 4 dp)}$$

$$\text{b) } e^{2x} \log y = 3$$

$$(e^{2x}) \left(\frac{1}{y} \frac{dy}{dx} \right) + (\log y) (2e^{2x}) = 0$$

$$\frac{e^{2x}}{y} \frac{dy}{dx} = -2e^{2x} \log y$$

$$\underline{\underline{\frac{dy}{dx} = -2y \log y}}$$

$$\text{c) } \frac{AR}{RB} = \frac{AQ}{QC} = \frac{1}{1} \text{ (given)}$$

$\therefore RQ \parallel BC$ (ratio of transversals are =)

$PQ \parallel AB$ (by similar method)

$\therefore RPQ$ is parallelogram (opposite sides are $\parallel$)

(ii) $\angle QRP = \angle QCP$ (opposite $\angle$'s in $\parallel$ gram =)

$\angle QRP = \angle QXC$ ($\angle$'s in same segment)

$\therefore \angle QCP = \angle QXC$

Thus $\triangle QXC$ is isosceles ($2 = \angle$'s)

(iii) $QC = QY$ (given)
 $QC = QX$ (= Sides in isosceles Δ)

$\therefore AC$ is a diameter of the circumcircle of ΔAXC , with Q being the centre.

$\angle AXC = 90^\circ$ ($\angle$ in semicircle)
 ie $AX \perp BC$

Question 8

a) i) Ways = $8!$
 = 40320

ii) No two M together

W^x W^x W^x W^x
 x x x x
 W^x W^x

Ways = $4! \times {}^5P_4$
 = 2880

iii) $P(M_A W_A \text{ together})$

= $\frac{4! \times 2 \times {}^4P_3}{2880}$

= $\frac{2}{5}$

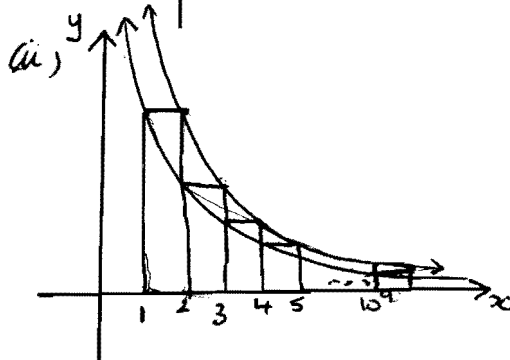
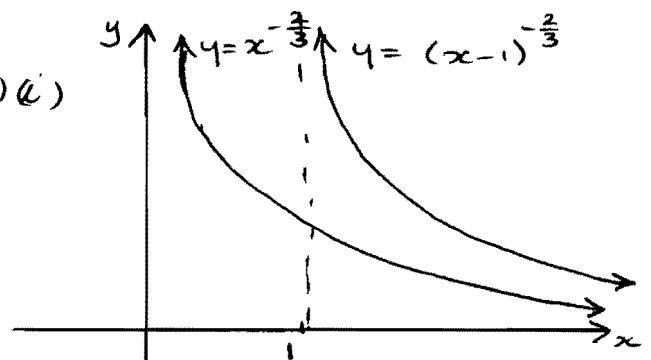
b) $x^3 + ax^2 + bx + c = 0$
 roots $\alpha, \beta, \alpha + \beta$

$2\alpha + 2\beta = -a$ $\alpha\beta + \alpha^2 + \alpha\beta + \alpha\beta + \beta^2 = b$
 $\alpha + \beta = -\frac{a}{2}$ $\alpha^2 + 3\alpha\beta + \beta^2 = b$
 $(\alpha + \beta)^2 + \alpha\beta = b$

$\alpha\beta(\alpha + \beta) = -c$
 $\alpha\beta(-\frac{a}{2}) = -c$
 $\alpha\beta = \frac{2c}{a}$

$(\alpha + \beta)^2 + \alpha\beta = b$
 $\frac{a^2}{4} + \frac{2c}{a} = b$
 $a^3 + 8c = 4ab$
 $a^3 - 4ab + 8c = 0$

c) i)



$\int_1^{10^9+1} x^{-\frac{2}{3}} dx < S < 1 + \int_1^{10^9+1} (x-1)^{-\frac{2}{3}} dx$
 $3 \left[x^{\frac{1}{3}} \right]_1^{10^9+1} < S < 1 + 3 \left[(x-1)^{\frac{1}{3}} \right]_1^{10^9+1}$
 $3 \left[(10^9+1)^{\frac{1}{3}} - 1 \right] < S < 1 + 3 \left[10^3 - 1 \right]$

$2997.000001 < S < 2998$

$\therefore S$ lies between 2997 and 2998.